PVSYST
PHOTOVOLTAIC
SOFTWARE

PVsystCLI TRIAL

PVsyst - Simulation report

Grid-Connected System

Project: Residential DEMO

Variant: First simulation

No 3D scene defined, no shadings

System power: 9.00 kWp

MCP_E2E_Site - Colombia

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Author

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VC0, Simulation date:
11/10/26 01:57
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Project summary

Geographical Site
MCP_E2E_Site
Colombia

Situation

Latitude	7.0700 °(N)
Longitude	-73.8400 °(W)
Altitude	0 m
Time zone	UTC-5

Project settings
Albedo 0.20

Weather data
MCP_E2E_Site
NASA SSE grid cell 7/-74 - Synthetic

System summary

Grid-Connected System
No 3D scene defined, no shadings

Orientation #1
Fixed plane
Tilt/Azimuth 10 / 0 °

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	30 units
Pnom total	9.00 kWp

Inverters

Nb. of units	1 unit
Total power	7.5 kWac
Pnom ratio	1.20

Results summary

Simulation step Hourly

Produced Energy	14299 kWh/year	Specific production	1589 kWh/kWp/year	Perf. Ratio PR	83.68 %

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

Simulation step
Hourly

Orientation #1
Fixed plane
Tilt/Azimuth 10 / 0 °

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon
Free Horizon

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	Generic
Model	Mono 300 Wp 60 cells
(Original PVsyst database)	
Unit Nom. Power	300 Wp
Number of PV modules	30 units
Nominal (STC)	9.00 kWp
Modules	2 units x 15 In series
At operating cond. (50°C)	
Pmpp	8.10 kWp
U mpp	427 V
I mpp	19 A

Inverter

Manufacturer	Generic
Model	7.5 kWac inverter
(Original PVsyst database)	
Unit Nom. Power	7.50 kWac
Number of inverters	1 unit
Total power	7.5 kWac
Operating voltage	150-750 V
Max. power (=>25°C)	8.00 kWac
Pnom ratio (DC:AC)	1.20
Power sharing within this inverter	

Total PV power

Nominal (STC)	9 kWp
Total	30 modules
Module area	48.8 m²
Cell area	44.2 m²

Total inverter power

Total power	7.5 kWac
Max. power	8 kWac
Number of inverters	1 unit
Pnom ratio	1.20

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 378 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss
Loss Fraction -0.75 %

Module mismatch losses
Loss Fraction 2.00 % at MPP

IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

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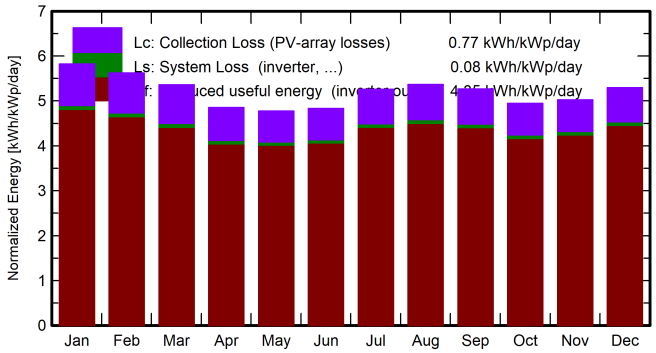
Main results

Simulation step
Hourly

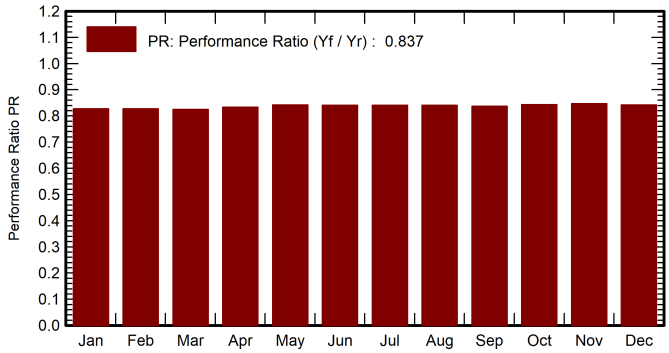
System Production

Produced Energy	14299 kWh/year	Specific production	1589 kWh/kWp/year
		Perf. Ratio PR	83.68 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

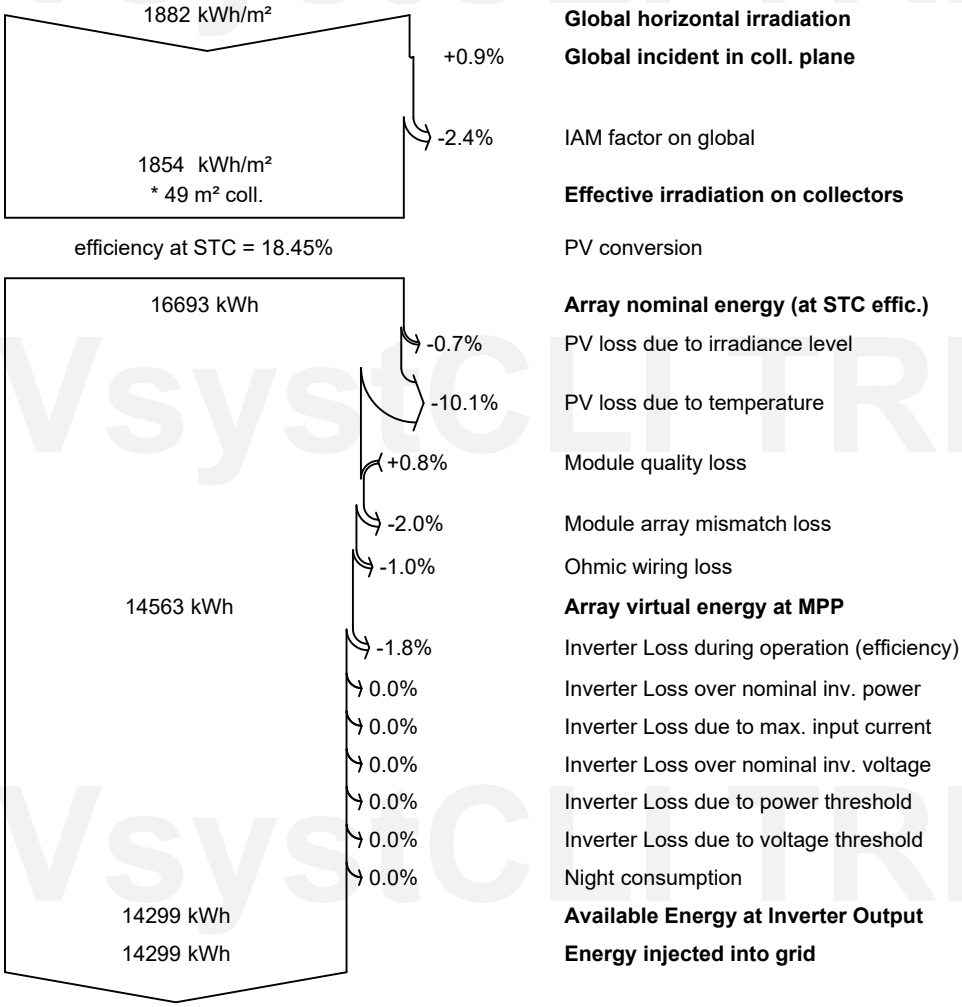
	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m²	kWh/m²	°C	kWh/m²	kWh/m²	kWh	kWh	ratio
January	165.5	57.62	23.30	180.5	176.6	1369	1344	0.827
February	149.5	61.40	24.70	157.5	154.2	1195	1173	0.828
March	163.7	69.43	24.50	166.2	162.5	1256	1233	0.824
April	148.8	70.02	23.70	145.7	142.1	1114	1093	0.833
May	155.3	74.30	23.20	147.9	144.1	1142	1121	0.842
June	154.8	68.90	23.10	145.1	140.9	1118	1098	0.841
July	172.7	76.72	23.40	163.0	158.8	1255	1233	0.841
August	172.0	78.24	23.60	166.3	162.3	1280	1258	0.840
September	158.1	71.52	23.00	158.1	154.6	1213	1191	0.837
October	148.8	77.28	22.40	153.4	150.0	1185	1163	0.843
November	141.6	71.29	22.30	150.7	147.4	1168	1148	0.846
December	150.7	66.21	22.40	164.3	160.5	1268	1244	0.842
Year	1881.5	842.93	23.29	1898.7	1853.9	14563	14299	0.837

Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

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Loss diagram



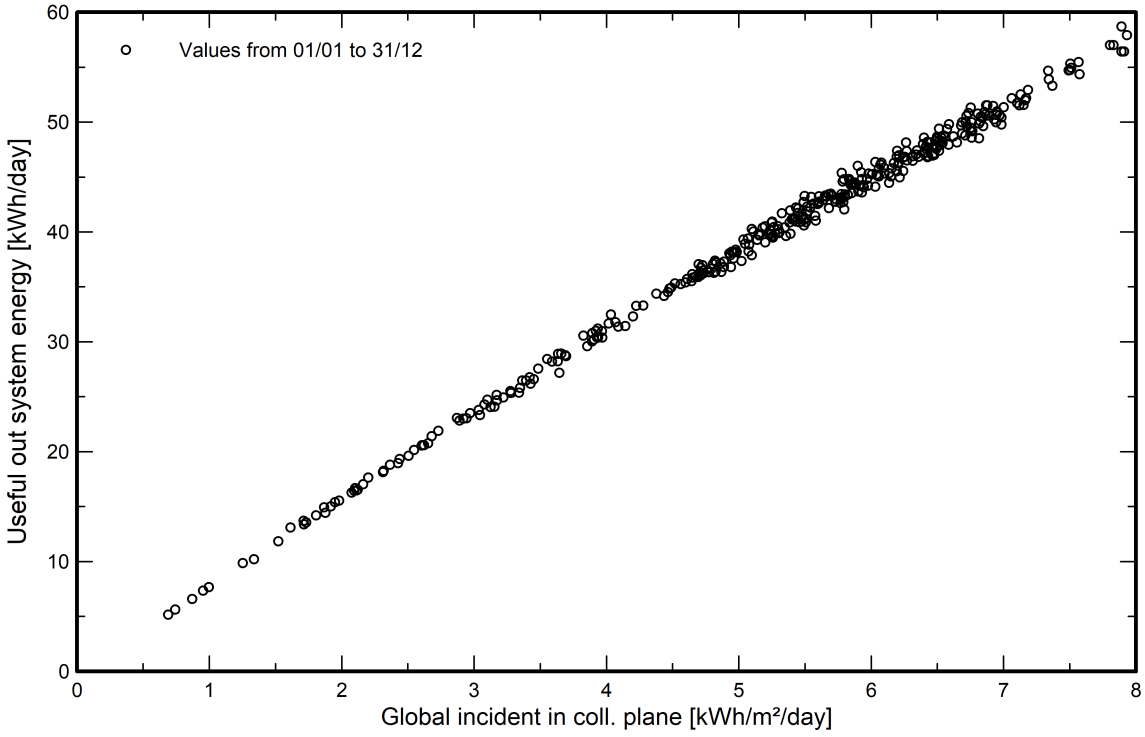
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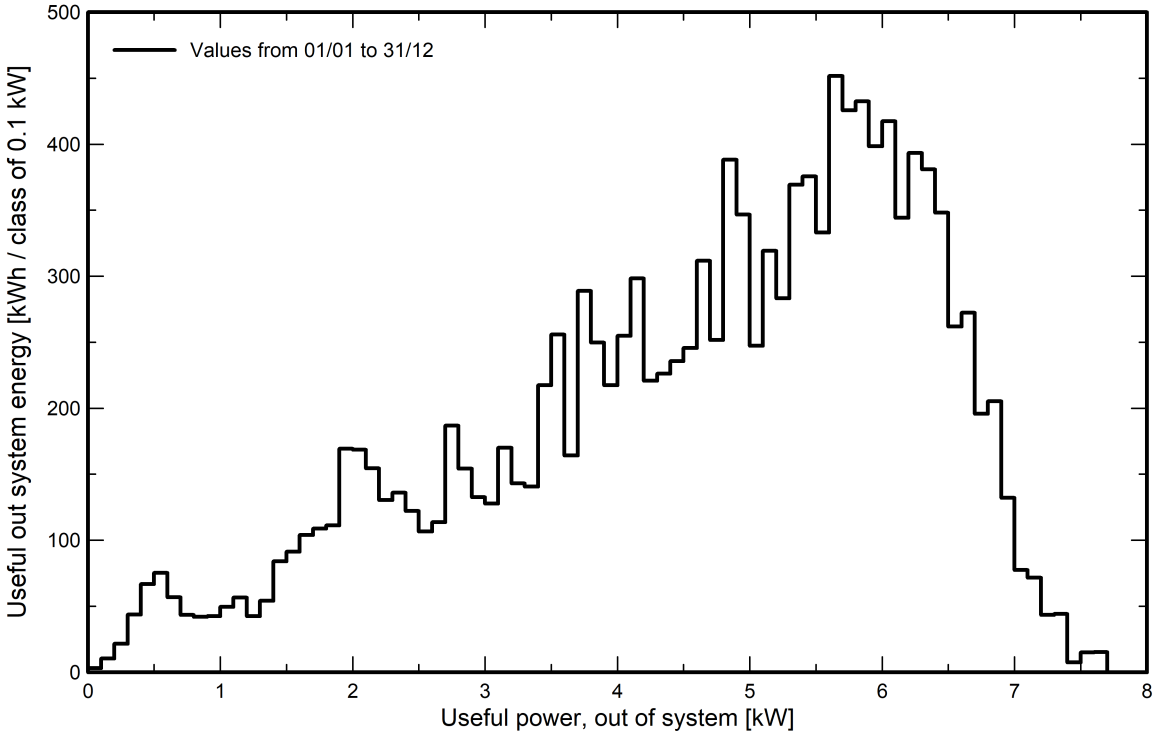
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Predef. graphs

Daily Input/Output diagram

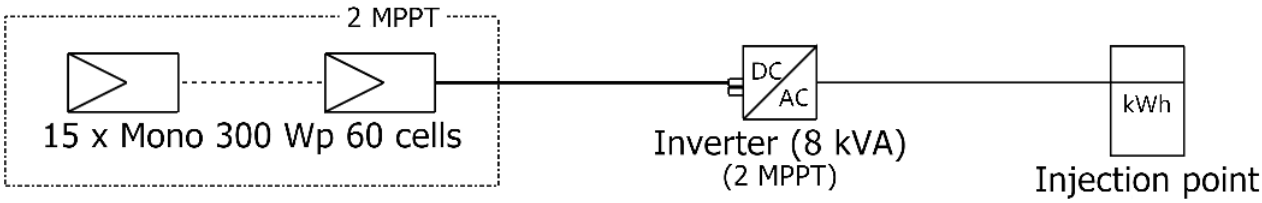


System Output Power Distribution



Single-line diagram

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PV module	Mono 300 Wp 60 cells
Inverter	7.5 kWac inverter
String	15 x Mono 300 Wp 60 cells

	Residential DEMO	
	VC0 : First simulation	11/10/26